

YouTube Trending Video Analytics

Project Report | Submitted by: Trisha N

1. Introduction

This project analyzes trending video data across regions to uncover patterns in popular categories, audience sentiment, engagement and trending duration. It follows the internship brief for YouTube Trending Video Analytics.

2. Abstract

The prepared dataset contains 1,500 unique videos across five regions and eight categories. Data cleaning removed duplicate video IDs and handled missing comments. Category and region summaries, sentiment scores, engagement rates, monthly trends and trending duration were analyzed. The clean dataset and dashboard-ready workbook support further Tableau visualization.

3. Tools Used

Python (Pandas, Matplotlib), SQL and Excel. The internship brief also specifies Tableau for the dashboard; the clean CSV and Excel workbook are structured for direct Tableau import.

4. Steps Involved

- Import and inspect the trending-video dataset.
- Remove duplicate video records and handle missing values.
- Standardize category, region and title fields.
- Calculate engagement rate and summarize average/total views.
- Rank categories by average views using SQL and Python.
- Compare regions and analyze sentiment by category.
- Study monthly trends and trending duration.
- Prepare dashboard-ready tables and visualizations.

5. Key Findings

In this prepared dataset, Sports has the highest average views among categories, while CA has the highest average views among regions. The correlation between trending duration and views is 0.01; this is an association and should not be interpreted as proof that longer trending duration causes higher views.

6. Conclusion

The project provides a structured data story around video popularity, engagement, sentiment and regional behavior. These insights can support content-performance analysis and can be extended in Tableau with filters for region, category and time.